

Microlocal Morse Theory and Truncated Microsupport

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Preface

In these notes, We study microlocal Morse theory, which is presented, for instance, in [GKS12]. However, we replace the notion of microsupports with that of truncated microsupprts, introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KMS03].

1 Notations

We consider a commutative unital ring $\mathbf{k}$ of finite global dimension (e.g. $\mathbf{Z}$ or a field). Let X be a real analytic manifold. We denote the cotangent bundle of X by $\pi: T^*X \rightarrow X$. We denote by $D^b(\mathbf{k}_X)$ the bounded derived category of sheaves of $\mathbf{k}$ -modules on X .

2 Truncated Microsupport

Definition 2.1. Let $F \in D^b(\mathbf{k}_X)$. The closed conic subset $SS_k(F)$ of T^*X is defined by: $p \notin SS_k(F)$ if and only if there exists an open conic neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1 -function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$(2.1) \quad H_{\{\varphi \geq 0\}}^j(F)_x = 0 \quad \text{for any } j \leq k.$$

2.1 Functorial Properties

Proposition 2.2. Let $f: X \rightarrow Y$ be a morphism of real analytic manifolds and let $F \in D^b(\mathbf{k}_X)$ such that f is proper on $\text{supp}(F)$. Then for any $k \in \mathbf{Z}$,

$$(2.2) \quad SS_k(Rf_*F) \subset f_\pi f_d^{-1}(SS_k(F)).$$

The equality holds in case f is a closed embedding.

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3 Lemmas from [KS90]

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary.

For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$(3.1) \quad E_n^\bullet: \cdots \rightarrow M_n^{j-1} \rightarrow M_n^j \rightarrow M_n^{j+1} \rightarrow \cdots,$$

and projective limit

$$(3.2) \quad E_\infty^\bullet: \cdots \rightarrow M_\infty^{j-1} \rightarrow M_\infty^j \rightarrow M_\infty^{j+1} \rightarrow \cdots.$$

Denote by

$$(3.3) \quad \Phi_k: H^k(E_\infty^\bullet) \rightarrow \varprojlim_n H^k(E_n^\bullet)$$

the natural morphism.

Proposition 3.1 ([KS90, Proposition 1.12.4]). Assume that for each $j \in \mathbf{Z}$, the system $(M_n^j)_n$ satisfies the ML condition. Then

- (a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,
- (b) if moreover, for a given i the system $(H^{i-1}(E_n^\bullet))_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 3.2 ([KS90, Proposition 1.12.6]). Let $(X_s, \rho_{s,t}: X_t \rightarrow X_s)_{(s, (s \leq t))}$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s: X_s \rightarrow \varprojlim_{r < s} X_r$$

and

$$\mu_s: \varinjlim_{t > s} X_t \rightarrow X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0, s_1} ($s_0 \leq s_1$) are injective (resp. surjective).

4 Results

In this section, let M be a manifold.¹

¹Can M be of class C^∞ ?

Theorem 4.1. Let F in $D^b(\mathbf{k}_M)$, and $\psi: M \rightarrow \mathbf{R}$ be a real-valued function of class C^1 , proper on $\text{supp}(F)$. Let $a < b$ be real numbers. We assume $d\psi(x) \notin \text{SS}_k(F)$ for $a \leq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t := \psi^{-1}(]-\infty, t])$. Then, the restriction morphisms

$$H^j(M_b; F) \rightarrow H^j(M_a; F)$$

are isomorphic for each $j < k$.²

Lemma 4.2. Let $-\infty < a < b < +\infty$ be real numbers and $G \in D^b(\mathbf{R})$. Let k be an integer. If

$$\left(H^j_{[t, +\infty[}(G) \right)_t \cong 0 \quad \text{for all } t \in [a, b[, \text{ and all } j \leq k.$$

Then one has

$$H^j(]-\infty, b[; G) \xrightarrow{\sim} H^j(]-\infty, a[; G)$$

for all $j < k$.³

Proof. Let $t \in [a, b[$. We will denote by I_t the open interval $]-\infty, t[$ and denote by Z_t the closed interval $]-\infty, t[(= \bar{I}_t)$. By applying the functor $(\cdot)_{\bar{I}_t}$ to the distinguished triangle

$$R\Gamma_{\mathbf{R}-I_t}(G) \rightarrow G \rightarrow R\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } D^b(\mathbf{k}_{\mathbf{R}}),$$

we have

$$(R\Gamma_{[t, +\infty[}(G))_{\{t\}} \rightarrow G_{Z_t} \rightarrow R\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } D^b(\mathbf{k}_{\mathbf{R}}).$$

Applying the functor $R\Gamma(\mathbf{R}; \cdot)$ to this, we have

$$(4.1) \quad (R\Gamma_{[t, +\infty[}(G))_t \rightarrow R\Gamma(Z_t; G) \rightarrow R\Gamma(I_t; G) \xrightarrow{+1} \quad \text{in } D^b(\mathbf{k}).$$

Then consider the following conditions for $j \in \mathbf{Z}$, $s \in [a, b[$, and $t \in]a, b[$:

$$\begin{aligned} (a)_s^j & \quad \varinjlim_{t>s} H^j(I_t; G) \xrightarrow{\sim} H^j(I_s; G), \\ (b)_t^j & \quad \varprojlim_{s<t} H^j(I_t; G) \xleftarrow{\sim} H^j(I_s; G). \end{aligned}$$

By hypothesis, it follows from (4.1) that $H^j(Z_t; G) \xrightarrow{\sim} H^j(I_t; G)$ for $t \in [a, b[$.⁴ Since $\varinjlim_{t>s} H^j(I_t; G) \cong H^j(Z_s; G)$, $(a)_s^j$ holds for any $j < k$ and $s \in [a, b[$.⁵

Moreover, since we assume G is a bounded complex, $(b)_t^j$ holds for $j \ll 0$ and

²For $j = k$, this morphism might be injective.

³For k , this morphism might be injective. (not written in this draft.)

⁴Similarly, we have injections $H^k(Z_t; G) \hookrightarrow H^k(I_t; G)$ for $t \in [a, b[$.

⁵Similarly,

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(Z_s; G)$$

gives injections

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(I_s; G)$$

for all $s \in [a, b[$.

for all $t \in]a, b]$. Let us denote such j by j_0 . Let us argue by induction on j . Assume $(b)_t^j$ holds for all $t \in]a, b[$ and all $j \leq j_0$. Applying Proposition 3.2, we find the isomorphisms

$$(4.2) \quad H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j \leq j_0 \text{ and all } a < s \leq t < b.$$

Let $G \xrightarrow{\sim} G^\bullet$ be a flabby resolution. Let $t \in]a, b]$. Consider a complex

$$E_n^\bullet = \Gamma(I_{t-\frac{1}{n}}; G^\bullet).$$

Since $G^\bullet$ is flabby, we have $E_n^j \cong E_\infty^j$ and the projective system $(E_n^j)_n$ satisfies the ML condition for all $j \in \mathbf{Z}$.

By (4.2), the projective system $(H^{j_0}(E_n^\bullet))_n$ satisfies the ML condition (because we have $a < t - \frac{1}{n} < t$ for $n \gg 0$). Applying Proposition 3.1 to $(E_n^\bullet)_n$, it follows that $(b)_t^{j_0+1}$ holds for $j_0 + 1$ and the induction proceeds. More precisely, we have

$$\begin{aligned} H^{j_0+1}(I_t; G^\bullet) &\xrightarrow{\sim} \varprojlim_{\Phi_{j_0+1}} H^{j_0+1}(I_{t-\frac{1}{n}}; G^\bullet) \\ &\cong \varprojlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G) \\ &\cong \varprojlim_{s < t} H^{j_0+1}(I_s; G). \end{aligned}$$

Again by Proposition 3.2, we get the isomorphisms

$$H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j < k \text{ and all } a < s \leq t \leq b,$$

and injections

$$H^j(I_s; G) \hookrightarrow H^j(I_t; G)$$

for $j = k$ and for all $a < s \leq t \leq b$.

Finally, using (4.1) and the hypothesis, we have the isomorphisms

$$H^j(I_a; G) \xleftarrow{\sim} H^j(Z_a; G) \xrightarrow{\sim} H^j(I_s; G) \quad \text{for all } j < k \text{ and all } a < s \leq b.$$

Similarly, for $j = k$, we have

$$H^k(I_b; G) \cong \varinjlim_{t > s} H^k(I_t; G) \hookrightarrow H^k(I_s; G) \hookrightarrow H^k(I_a; G).$$

□

Proof of Theorem 4.1. First, we have

$$H^j(M_t; F) \cong H^j \mathrm{R}\Gamma(]-\infty, t[; \mathrm{R}\psi_* F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leq \psi(x) < b$, and by (2.2), the formula $\mathrm{SS}_k(\mathrm{R}\psi_* F) \subset \psi_\pi \psi_d^{-1} \mathrm{SS}_k(F)$ holds. Hence we have

$$(H^j \mathrm{R}\Gamma_{[\psi(x), +\infty[} \mathrm{R}\psi_* F)_{\psi(x)} = 0$$

for each $j \leq k$. Then we can apply Lemma 4.2 and have

$$H^j(]-\infty, b[; \mathrm{R}\psi_* F) \xrightarrow{\sim} H^j(]-\infty, a[; \mathrm{R}\psi_* F)$$

for all $j < k$. Then we can conclude that

$$H^j(M_b; F) \xrightarrow{\sim} H^j(M_a; F)$$

for all $j < k$.

□

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